Lösungen · Trigonometrie

1. $\mathbf{a = \frac{7{,}4}{\sin 52°} \approx 9{,}4\ m}$ ..... $\sin 52° = \frac{7{,}4}{a}$

2. $\mathbf{BF \approx 93{,}0\ cm}$ ..... $\sin 45° = BF : 131{,}5$
$\mathbf{BC = BF : \cos 65° \approx 220{,}0\ cm}$ ..... $\cos 65° = BF : BC$

3. **100 und 6**
ja ..... $\frac{16}{170} = \frac{8}{85} \approx 9{,}4\ \% > 6\ \%$

4. $\mathbf{AC = \frac{h}{\sin 22°} \approx 78{,}1\ cm}$ ..... $h = \sqrt{32^2 - 13^2} = 3\sqrt{95}$; $\sin 22° = h : AC$

5. **a) AB ≈ 1228,7 m** ..... $\sin 55° = AB : 1500$

b) AD ≈ 860,4 m ..... $\cos 55° = AD : 1500$

6. $\mathbf{PB = \frac{1{,}20}{\tan 34{,}9°} \approx 1{,}72\ m}$ ..... $\tan 34{,}9° = 1{,}20 : PB$

7. **Berg = 57,9 m; D = 69,9 m** ..... $\sin 30° = BC : 112{,}8$; Berg = BC + 1,5; D = 57,9 + 12,0

8. **BD ≈ 502 m** ..... $\sin 54° = AD : 920$; $\tan 34° = BD : AD$